

Fast, Efficient and Reliable Pattern Scanning and Processing

Introducing Awk, a language that has built-in features to recognise patterns and manipulate them with ease.

 Whether the need is to provide some useful information to customers or the management, monitor a system's activity, extract certain lines from a file, or to automate certain tasks, a systems administrator has to often deal with text files. Carrying out these tasks usually requires parsing or manipulating text files, which might be as small as a few lines or as huge as a few gigabytes. Although there are various ways of doing this in *nix systems, most are either too trivial or so complex that one has to write several lines of code to accomplish a small task. However, Awk, with its built-in features to recognise patterns and its ability to manipulate them easily, is one of the fastest, most efficient and reliable tools. It combines the best of both worlds by providing the ease of use of grep-like tools, and the features and efficiency of a programming language. Its C-like syntax is

very easy to learn.

Awk was created by Alfred V. Aho, Peter J. Weinberger and Brian W. Kernighan (one of the creators of C language). It gets its name from the initials of its creators. Awk has evolved over the years and the current version is usually called Nawk for 'New Awk'. Gawk, the GNU version of Awk, supports the new features as well as some GNU-specific extensions. Gawk is included with the various Linux distributions and is used widely in several start-up scripts. Usually, in most Linux distributions, awk is a symbolic link to the gawk utility.

Records and fields

An Awk program divides the input file(s) into records and fields. The input file is divided into records based on a Record Separator (RS) variable. By default, records are separated by a newline character. However, an RS can be any single character or a regular expression. For example, setting RS=" \$"